

Member ID: _____

Time: _____

Rank: _____



COMPUTER SECURITY (320)

STATE 2023

TOTAL POINTS (50 @ 2 points each) _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. What type of network threat is a virus?
 - A. Malware
 - B. Phishing
 - C. Spam
 - D. Denial of Service (DoS)

2. What type of network threat is a fake email or website that is designed to trick you into giving away personal information?
 - A. Malware
 - B. Phishing
 - C. Spam
 - D. Denial of Service (DoS)

3. What type of network threat is a flood of unwanted or unsolicited emails?
 - A. Malware
 - B. Phishing
 - C. Spam
 - D. Denial of Service (DoS)

4. What type of network threat is an attack that aims to make a network or website unavailable to users?
 - A. Malware
 - B. Phishing
 - C. Spam
 - D. Denial of Service (DoS)

5. What type of network threat is a program or script that is designed to harm or damage a computer or network?
 - A. Malware
 - B. Phishing
 - C. Spam
 - D. Denial of Service (DoS)

6. Which type of biometric security is considered the most secure?
 - A. Facial recognition
 - B. Voice recognition
 - C. Fingerprint recognition
 - D. Multi-factor biometric authentication

7. What type of SSL certificate is used to secure a single domain?
 - A. Wildcard SSL
 - B. Extended Validation SSL
 - C. Single Domain SSL
 - D. Multi-Domain SSL

8. Which type of SSL certificate is used to secure multiple subdomains under a single domain?
 - A. Wildcard SSL
 - B. Extended Validation SSL
 - C. Single Domain SSL
 - D. Multi-Domain SSL

9. What type of SSL certificate is used to provide the highest level of security and validation for a website?
 - A. Wildcard SSL
 - B. Extended Validation SSL
 - C. Single Domain SSL
 - D. Multi-Domain SSL

10. How long is the typical lifespan of an SSL certificate?
 - A. 1 year
 - B. 2 years
 - C. 3 years
 - D. 5 years

11. What type of SSL certificate is used to secure multiple different domains under a single certificate?
 - A. Wildcard SSL
 - B. Extended Validation SSL
 - C. Single Domain SSL
 - D. Multi-Domain SSL

12. What is the purpose of the Security Options setting in the Windows Security Policy?
 - A. To set the password policy for user accounts
 - B. To configure advanced security settings for the operating system
 - C. To control access to network resources
 - D. To manage user permissions for specific files and folders

13. Which of the following is not a built-in security group in Windows?
- A. Administrators
 - B. Users
 - C. Guests
 - D. Superusers
14. What is the purpose of the Audit Policy in the Windows Security Policy?
- A. To track user activity on the computer
 - B. To prevent unauthorized access to the system
 - C. To set the password policy for user accounts
 - D. To manage user permissions for specific files and folders
15. Which of the following is not a built-in user right in Windows?
- A. Logging on locally
 - B. Shutting down the system
 - C. Creating a new user account
 - D. Changing the local administrator account password
16. What is the purpose of the User Rights Assignment setting in the Windows Security Policy?
- A. To configure advanced security settings for the operating system
 - B. To control access to network resources
 - C. To assign specific user rights and permissions to individual users or groups
 - D. To set the password policy for user accounts
17. What is the primary protocol used for transmitting data over the internet?
- A. TCP
 - B. UDP
 - C. FTP
 - D. HTTP
18. What is the primary function of the IP (Internet Protocol) in the TCP/IP model?
- A. To ensure data is delivered to the correct destination
 - B. To ensure data is delivered in the correct order
 - C. To ensure data is delivered securely
 - D. To ensure data is compressed for faster transmission

19. What is the primary function of the TCP (Transmission Control Protocol) in the TCP/IP model?
- A. To ensure data is delivered to the correct destination
 - B. To ensure data is delivered in the correct order
 - C. To ensure data is delivered securely
 - D. To ensure data is compressed for faster transmission
20. What is the difference between TCP and UDP?
- A. TCP is connection oriented and UDP is connectionless
 - B. TCP is faster than UDP
 - C. TCP is more secure than UDP
 - D. TCP is better for streaming video, while UDP is better for gaming
21. What is the purpose of the DNS (Domain Name System) in the TCP/IP model?
- A. To translate domain names into IP addresses
 - B. To ensure data is delivered to the correct destination
 - C. To ensure data is delivered in the correct order
 - D. To ensure data is delivered securely
22. What is the purpose of a risk matrix in risk analysis?
- A. To identify potential hazards
 - B. To assess the likelihood and impact of each hazard
 - C. To prioritize risks based on their likelihood and impact
 - D. To implement controls to mitigate the risk
23. What is the purpose of a risk register?
- A. To document all identified risks
 - B. To assign ownership of risks
 - C. To track the status of risk management actions
 - D. All of the other answers are correct
24. What is SELinux and why is it important for hardening a Linux system?
- A. SELinux is a type of firewall that provides network security
 - B. SELinux is a security enhancement to Linux that provides mandatory access control
 - C. SELinux is a type of antivirus software that detects and removes malware
 - D. SELinux is a type of encryption software that protects sensitive data

25. What is the purpose of the `chmod` command in hardening a Linux system?
- A. To set permissions for files and directories to restrict access
 - B. To update software packages on a Linux system
 - C. To encrypt sensitive data on a Linux system
 - D. To run a scan for malware on a Linux system
26. Which file should be used to configure user access and permissions on a Linux system?
- A. `/etc/passwd`
 - B. `/etc/shadow`
 - C. `/etc/sudoers`
 - D. `/etc/group`
27. What is the command used to view the current status of the firewall on a Linux system?
- A. `firewall-status`
 - B. `iptables -L`
 - C. `firewall-config`
 - D. `ufw status`
28. Which command is used to enable the SSH service on a Linux system?
- A. `ssh-on`
 - B. `service ssh start`
 - C. `systemctl start ssh`
 - D. `enable-ssh`
29. How can you set a password for the root user on a Linux system?
- A. `passwd root`
 - B. `set-root-password`
 - C. `usermod -p root`
 - D. `change-root-password`
30. Which group policy setting can be used to disable the use of USB storage devices on a Windows system?
- A. Device Installation Restrictions
 - B. Removable Storage Access
 - C. Security Options
 - D. System Services
31. What is the PowerShell command to enable BitLocker encryption on a Windows system?
- A. `Enable-BitLocker`
 - B. `Start-BitLockerEncryption`
 - C. `Set-BitLocker -EncryptionEnabled $True`
 - D. `manage-bde -on c:`

32. What type of wireless security protocol uses Temporal Key Integrity Protocol (TKIP) to encrypt data?
- A. WPA
 - B. WPA2
 - C. WPA3
 - D. WEP
33. What type of wireless security protocol uses Advanced Encryption Standard (AES) to encrypt data?
- A. WPA
 - B. WPA2
 - C. WPA3
 - D. WEP
34. What is the Caesar shift of the word “encryption” with a shift value of -3?
- A. hqfwrjudsk
 - B. irqfswbqwj
 - C. ajynulpekj
 - D. bkbmqiwtqj
35. The decimal number 12 is represented in the hexadecimal system as _____.
- A. A
 - B. B
 - C. C
 - D. D
36. Which of the following protocols can be used to encrypt network communication and secure data transmission over a public network?
- A. FTP
 - B. Telnet
 - C. HTTPS
 - D. SMTP
37. How can you use your smart phone to login to windows 11?
- A. Microsoft Authenticator
 - B. Microsoft Hello
 - C. Microsoft Cloud
 - D. Microsoft Pin

38. What was added as an added security requirement for Windows 11 installations?
- A. TPM
 - B. Bitlocker
 - C. Secure Key Subscriptions
 - D. SmartScan
39. Which of the following security features is new in Microsoft Windows 11?
- A. Windows Hello
 - B. Windows SmartScreen
 - C. Windows Sandbox
 - D. Windows Defender Exploit Guard
40. What are the required records needed to generate a SSL certificate?
- A. CSR, request, response, Contact Details
 - B. CSR, Public Key, Private Key, Contact Details
 - C. CSR, Sign File, KeyGen, Contact Details
 - D. CSR, Sign File, .TXT file, Contact Details
41. Which method does not secure the password and exposes it to plain text to send usernames and passwords?
- A. Hash
 - B. Salt
 - C. Encryption
 - D. PAP
42. What type of attack would normally go hand in hand with Social Engineering?
- A. Badgering
 - B. Honey badgering
 - C. Trojan phishing
 - D. Phishing
43. What type of authentication uses a renewable session ticket?
- A. NTLM
 - B. Kerberos
 - C. Chap
 - D. EAP

44. What Windows utility keeps settings for applications, devices, software, users' settings and more?
- A. Group policy
 - B. Registry
 - C. Users and groups
 - D. Services
45. What is ARP spoofing is also known as?
- A. ARP honeypot
 - B. ARP botnet
 - C. DNS poisoning
 - D. ARP poisoning
46. What is a two-party agreement about the accepted use of corporate network or internet is known as?
- A. Firewall policy
 - B. Access policy
 - C. Acceptable use policy
 - D. Network use policy
47. A database server needs to be accessed by your Internet web servers. Where do you place it?
- A. DMZ
 - B. Internal network
 - C. Internet
 - D. VPN
48. What port would you block to prevent users from using the Time protocol?
- A. 80
 - B. 443
 - C. 25
 - D. 123
49. Which is not a Malware removal tool?
- A. McAfee
 - B. Malwarebytes
 - C. AVG
 - D. Wormee
50. Which of the following is a feature of Windows Defender that can help protect your computer from malicious software?
- A. Parental controls
 - B. Automatic updates
 - C. Real-time protection
 - D. Disk defragmentation**